Pionless NLO and P-waves

Lorenzo Contessi, Alessandro Lovato

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Definition and conventions

$$
\begin{array}{cccccc}
(p') & | & & | & (-p') \\
 & N & & N & \\
 & | & -- & | & \\
 & N & & N & \\
(p) & | & & | & (-p)
\end{array}
\tag{1}
$$

$$
\begin{array}{ll}
\vec{q} = \vec{p'} - \vec{p} & \vec{k} = \vec{p'} + \vec{p} \\
\mathcal{P}_{(s,I)} & \text{Projector in the total spin = s/2 and total isospin = I/2 channel} \\
\mathcal{P}_{(0,0)} = & \frac{1}{4}(1 - \sigma_{12})\frac{1}{4}(1 - \tau_{12}) \\
\mathcal{P}_{(0,2)} = & \frac{1}{4}(1 - \sigma_{12})\frac{1}{4}(3 + \tau_{12}) \\
\mathcal{P}_{(2,0)} = & \frac{1}{4}(3 + \sigma_{12})\frac{1}{4}(1 - \tau_{12}) \\
\mathcal{P}_{(2,2)} = & \frac{1}{4}(3 + \sigma_{12})\frac{1}{4}(3 + \tau_{12}) \\
\mathcal{P}_{(*,0)} = & \frac{1}{4}(1 - \tau_{12}) \\
\mathcal{P}_{(*,2)} = & \frac{1}{4}(3 + \tau_{12}) \\
\mathcal{P}^{123}_{(1,1)} = & \frac{1}{36}(3 - \tau_{12} + \tau_{23} + \tau_{31})(3 - \sigma_{12} + \sigma_{23} + \sigma_{31})
\end{array}
\tag{2}
$$

1 Introduction

The goal of this notes is to implement a simple but "theoretical-controlled" version of the EFT($\not\pi$) at NLO. Over the last months, we tried to implement the non-local NLO of EFT($\not\pi$) perturbatively in quantum Monte Carlo without conclusive results. Here we employ the Fierz transformations with the aim of developing a "minimally non local" version of the NLO contributions suitable to be employed in QMC. Indeed, terms depending on $\vec{k}$, although in principle calculable, are not ideal for QMC, as they are non-local. In the derivation of a local interaction we will specify the approximations that we are implicitly making and how and when they can be safely neglected. Its effect on S- and P- wave scattering will be thoroughly analyzed.

The version of the potential that we discuss in this notes has been proposed by Johannes a few years ago, based on Nir notes. Last summer Nir proposed to promote P-waves, but there were no practical ideas how to do that.

2 EFT($\not\pi$) potential

2.1 Two-body potential

The LO two-body potential in EFT($\not\pi$) is given by

$$V_{LO}^{2b} = f_\Lambda(q,k)\left(\beta_1\mathcal{P}_{(2,0)} + \beta_2\mathcal{P}_{(0,2)} + \beta_3\mathcal{P}_{(0,0)} + \beta_4\mathcal{P}_{(2,2)}\right) \tag{3}$$

In the naive-dimensional analysis, the NLO contributions to the potentials are

$$\begin{aligned} V_{NLO} = \; & f_\Lambda(q,k)[\\ & (\gamma_1 q^2\mathcal{P}_{(2,0)} + \gamma_2 q^2\mathcal{P}_{(0,2)} + \gamma_3 q^2\mathcal{P}_{(0,0)} + \gamma_4 q^2\mathcal{P}_{(2,2)})+ \\ & (\gamma_5 k^2\mathcal{P}_{(2,0)} + \gamma_6 k^2\mathcal{P}_{(0,2)} + \gamma_7 k^2\mathcal{P}_{(0,0)} + \gamma_8 k^2\mathcal{P}_{(2,2)})+ \\ & (\gamma_9(\sigma_1+\sigma_2)\cdot(q\times k)\mathcal{P}_{(*,0)} + \gamma_{10}(\sigma_1+\sigma_2)\cdot(q\times k)\mathcal{P}_{(*,2)}+ \\ & (\gamma_{11}(\sigma_1\cdot q)(\sigma_2\cdot q)\mathcal{P}_{(*,0)} + \gamma_{12}(\sigma_1\cdot q)(\sigma_2\cdot q)\mathcal{P}_{(*,2)})+ \\ & (\gamma_{13}(\sigma_1\cdot k)(\sigma_2\cdot k)\mathcal{P}_{(*,0)} + \gamma_{14}(\sigma_1\cdot k)(\sigma_2\cdot k)\mathcal{P}_{(*,2)})] \end{aligned} \tag{4}$$

The above expressions can be simplified if we assume that the regulator is such that $f_\Lambda(k,q) = f_\Lambda(q,k)$. In fact, the wave-function of a fermionic system is such that

$$(12)|\Psi\rangle = -|\Psi\rangle \tag{5}$$

where (12) exchanges particles 1 and 2. Therefore, the anti-symmetrized potential

$$V_{12}^{\mathrm{as}} = \frac{1}{2}\left[V_{12} - (12)V\right] \tag{6}$$

when applied to an anti-symmetric wave function is equivalent to the original potential

$$V_{12}^{\mathrm{as}}|\Psi\rangle \equiv V_{12}|\Psi\rangle\,. \tag{7}$$

Note that our QMC methods do not require to anti-symmetrize the potential, but this procedure will be used to make the potential "minimally non local". Besides the obvious identities $\vec{q}\overleftarrow{(12)} = \vec{k}$ and $\vec{k}\overleftarrow{(12)} = \vec{q}$, the following relations will prove to be useful

$$
\begin{aligned}
(12)\mathcal{P}_{(0,0)} &= \mathcal{P}_{(0,0)} \\
(12)\mathcal{P}_{(2,0)} &= -\mathcal{P}_{(2,0)} \\
(12)\mathcal{P}_{(0,2)} &= -\mathcal{P}_{(0,2)} \\
(12)\mathcal{P}_{(2,2)} &= \mathcal{P}_{(2,2)} \\
(12)(\sigma_1+\sigma_2)\cdot(q\times k)P_{(*,0)} &= (\sigma_1+\sigma_2)\cdot(q\times k)\mathcal{P}_{(*,0)} \\
(12)(\sigma_1+\sigma_2)\cdot(q\times k)\mathcal{P}_{(*,2)} &= -(\sigma_1+\sigma_2)\cdot(q\times k)\mathcal{P}_{(*,2)} \\
(12)(\sigma_1\cdot q)(\sigma_2\cdot q)\mathcal{P}_{(*,0)} &= -(\sigma_1\cdot k)(\sigma_2\cdot k)\mathcal{P}_{(*,0)} - 2k^2\mathcal{P}_{(0,0)} \\
(12)(\sigma_1\cdot q)(\sigma_2\cdot q)\mathcal{P}_{(*,2)} &= (\sigma_1\cdot k)(\sigma_2\cdot k)\mathcal{P}_{(*,2)} + 2k^2\mathcal{P}_{(0,2)} \\
(12)(\sigma_1\cdot k)(\sigma_2\cdot k)\mathcal{P}_{(*,0)} &= -(\sigma_1\cdot q)(\sigma_2\cdot q)\mathcal{P}_{(*,0)} - 2q^2\mathcal{P}_{(0,0)} \\
(12)(\sigma_1\cdot k)(\sigma_2\cdot k)\mathcal{P}_{(*,2)} &= (\sigma_1\cdot q)(\sigma_2\cdot q)\mathcal{P}_{(*,2)} + 2q^2\mathcal{P}_{(0,2)}\,.
\end{aligned}
\tag{8}
$$

The LO contribution is unambiguously determined by

$$
V_{\rm LO}^{\rm as} = f_\Lambda(q,k)(C_{20}\mathcal{P}_{(2,0)} + C_{02}\mathcal{P}_{(0,2)})
\tag{9}
$$

As for the NLO, let us first consider the $T=0$ channel

$$
\begin{aligned}
V_{\rm NLO}^{\rm as}(*,0) = f_\Lambda(q,k)\Big[&\frac{(\gamma_3-\gamma_7+2\gamma_{13})}{2}q^2\mathcal{P}_{(0,0)} + \frac{(\gamma_7-\gamma_3+2\gamma_{11})}{2}k^2\mathcal{P}_{(0,0)} \\
&+ \frac{(\gamma_1+\gamma_5)}{2}q^2\mathcal{P}_{(2,0)} + \frac{(\gamma_1+\gamma_5)}{2}k^2\mathcal{P}_{(2,0)} + \frac{(\gamma_{11}+\gamma_{13})}{2}(\sigma_1\cdot q)(\sigma_2\cdot q)\mathcal{P}_{(*,0)} \\
&+ \frac{(\gamma_{11}+\gamma_{13})}{2}(\sigma_1\cdot k)(\sigma_2\cdot k)\mathcal{P}_{(*,0)}\Big]
\end{aligned}
\tag{10}
$$

It is immediate to see that once anti-symmetrized

$$
V_{\rm NLO}(*,0) = f_\Lambda(q,k)\Big[E_{00}q^2\mathcal{P}_{(0,0)} + E_{20}q^2\mathcal{P}_{(2,0)} + E_{*0}(\sigma_1\cdot q)(\sigma_2\cdot q)\mathcal{P}_{(*,0)}\Big]
\tag{11}
$$

provides all the linear independent contributions of Eq. (10). The $T=1$ channel reads

$$
\begin{aligned}
V_{\rm NLO}^{\rm as}(*,1) = f_\Lambda(q,k)\Big[&\frac{(\gamma_2+\gamma_6-2\gamma_{14})}{2}q^2\mathcal{P}_{(0,2)} + \frac{(\gamma_2+\gamma_6-2\gamma_{12})}{2}k^2\mathcal{P}_{(0,2)} \\
&+ \frac{(\gamma_4-\gamma_8)}{2}q^2\mathcal{P}_{(2,2)} + \frac{(\gamma_8-\gamma_4)}{2}k^2\mathcal{P}_{(2,2)} + \frac{(\gamma_{12}-\gamma_{14})}{2}(\sigma_1\cdot q)(\sigma_2\cdot q)\mathcal{P}_{(*,2)} \\
&+ \frac{(\gamma_{14}-\gamma_{12})}{2}(\sigma_1\cdot k)(\sigma_2\cdot k)\mathcal{P}_{(*,2)} + \gamma_{10}(\sigma_1+\sigma_2)(q\times k)\mathcal{P}_{(*,2)}\Big]
\end{aligned}
\tag{12}
$$

Therefore, when anti-symmetrized

$$V_{\mathrm{NLO}}(*,2) = f_\Lambda(q,k)\Big[E_{02}q^2\mathcal{P}_{(0,2)} + E_{22}q^2\mathcal{P}_{(2,2)} \\ + E_{*2}(\sigma_1\cdot q)(\sigma_2\cdot q)\mathcal{P}_{(*,2)} + E_{LS}(\sigma_1+\sigma_2)(q\times k)\mathcal{P}_{(*,2)}\Big] \tag{13}$$

provides all the linear-independent terms of Eq. (12).

It has been argued that, because of the unnaturally scattering length in the two-body S-wave, NLO terms corresponding to γ_{3-4}, γ_{7-8}, and γ_{9-14} in Eq. (4) are demoted to N^{2}LO. In addition, since the term proportional to $(\sigma_1+\sigma_2)(q\times k)\mathcal{P}_{(*,0)}$ vanishes when anti-symmetrized, we can use it to remove the τ_{12} term in $\mathcal{P}_{(*,2)}$. Hence, redefining E_{LS}, we can consider

$$\begin{aligned}
V_{\mathrm{LO}} &= f_\Lambda(q,k)\left(C_{20}\mathcal{P}_{(2,0)} + C_{02}\mathcal{P}_{(0,2)}\right)\\
V_{\mathrm{NLO}} &= f_\Lambda(q,k)\left(E_{20}q^2\mathcal{P}_{(2,0)} + E_{02}q^2\mathcal{P}_{(0,2)}\right)\\
V_{\mathrm{N^2LO}} &= f_\Lambda(q,k)\left(E_{*0}(\sigma_1\cdot q)(\sigma_2\cdot q)\mathcal{P}_{(*,0)} + E_{*,2}(\sigma_1\cdot q)(\sigma_2\cdot q)\mathcal{P}_{(*,2)}\right.\\
&\quad\left. + iE_{\mathrm{LS}}/2(\sigma_1+\sigma_2)(q\times k) + E_{00}q^2\mathcal{P}_{(0,0)} + E_{22}q^2\mathcal{P}_{(2,2)}\right)
\end{aligned} \tag{14}$$

to be completely equivalent to the full ones when applied to anti-symmetric wave functions.[1] Note that, since the Fierz transformations mix NLO with N^{2}LO contributions, once $V_{\mathrm{N^2LO}}$ is included, appropriate N^{2}LO counterterms for E_{00} and E_{02} have to be included.

2.2 Local regulator

Since the exchanged momentum is the divergent quantity in the two-body bound and scattering states, a regulator that only depends upon the momentum transfer, $f_\Lambda(k,q) = f_\Lambda(q)$, suffices to regularize the potential. In addition, this particular choice of the regulator is convenient for QMC purposes, as it provides a local interaction. However, $f_\Lambda(q)$ brings about some issues when employing the Fierz transformation, essentially because $(12)f_\Lambda(q) = f_\Lambda(k)$. We choose $f_\Lambda(q) = e^{-q^2/\Lambda^2}$ that fulfills the requirement $\lim_{q\to 0} f_\Lambda(q) = 1$ for and $\lim_{q\to\infty} f_\Lambda(q) = 0$. When applying the exchange operator to this local regulator we get

$$(12)\exp\left(-\tfrac{q^2}{\Lambda^2}\right) = \exp\left(-\tfrac{k^2}{\Lambda^2}\right) = \exp\left(-\tfrac{q^2}{\Lambda^2}\right)g(k^2-q^2) \tag{15}$$

where $g(k^2-q^2) = \exp\left[-(k^2-q^2)/\Lambda^2\right] = \exp\left(-2p\cdot p'/\Lambda^2\right)$. Employing this cutoff function, the LO anti-symmetrized potential reads

$$V^{\mathrm{as}}_{\mathrm{LO}} = f_\Lambda(q)\left[\beta_1\frac{1+g(k^2-q^2)}{2}\mathcal{P}_{(2,0)} + \beta_2\frac{1+g(k^2-q^2)}{2}\mathcal{P}_{(0,2)} + \right.\\ \left.\beta_3\frac{1-g(k^2-q^2)}{2}\mathcal{P}_{(0,0)} + \beta_4\frac{1-g(k^2-q^2)}{2}\mathcal{P}_{(2,2)}\right]. \tag{16}$$

[1]Notice that at N^{2}LO additional S-wave contributions proportional to k^4 and q^4 should be included. They are not shown in Eq. 14 since they are not relevant for the present discussion.

Therefore, for finite Λ, the terms proportional to β_{3-4} do not vanish as in Eq. (9). The divergences arising from integrating the terms proportional to $g(k^2-q^2)$ can be reabsorbed in the low-energy constants β_{1-4} at leading order, as $\int d^3q f_\Lambda(q) g(k^2-q^2) \propto \Lambda^3$. Therefore, since

$$\lim_{\Lambda\to\infty} g(k^2-q^2) = 1\,. \tag{17}$$

we can set β_{3-4} to zero as their subleading contributions are provided by the corresponding NLO terms. It has to be noted that P-wave spuriosities disappear when all the particles of the systems are in relative S-wave. This analysis is supported by the empirical convergence in Λ of the theory at leading order.

Hence, we can write Eq. (14) as

$$\begin{aligned}
V_{\rm LO} &= f_\Lambda(q)\left(C_{20}\mathcal{P}_{(2,0)} + C_{02}\mathcal{P}_{(0,2)}\right) \\
V_{\rm NLO} &= f_\Lambda(q)\left(E_{20}q^2\mathcal{P}_{(2,0)} + E_{02}q^2\mathcal{P}_{(0,2)}\right) \\
V_{\rm N^2LO} &= f_\Lambda(q)\left(E_{*0}(\sigma_1\cdot q)(\sigma_2\cdot q)\mathcal{P}_{(*,0)} + E_{*2}(\sigma_1\cdot q)(\sigma_2\cdot q)\mathcal{P}_{(*,2)}\right. \\
&\quad \left. + iE_{\rm LS}/2(\sigma_1+\sigma_2)(q\times k) + E_{00}q^2\mathcal{P}_{(0,0)} + E_{22}q^2\mathcal{P}_{(2,2)}\right)
\end{aligned} \tag{18}$$

2.3 Coordinate-space interaction

The coordinate-space matrix elements of the potential are defined as

$$\begin{aligned}
V(\mathbf{r}',\mathbf{r}) &= \int \frac{d^3p}{(2\pi)^3}\frac{d^3p'}{(2\pi)^3}\langle\mathbf{r}'|\mathbf{p}'\rangle\langle\mathbf{p}'|V|\mathbf{p}\rangle\langle\mathbf{p}|\mathbf{r}\rangle \\
&= \int \frac{d^3p}{(2\pi)^3}\frac{d^3p'}{(2\pi)^3} e^{i\mathbf{p}'\cdot\mathbf{r}'}e^{-i\mathbf{p}\cdot\mathbf{r}}\,V(\mathbf{p}',\mathbf{p}) \\
&= \frac{1}{8}\int \frac{d^3k}{(2\pi)^3}\frac{d^3q}{(2\pi)^3} e^{i\mathbf{q}\cdot\mathbf{x}}e^{-i\mathbf{k}\cdot\mathbf{y}}\,V(\mathbf{k},\mathbf{q})
\end{aligned} \tag{19}$$

where $\mathbf{x} = (\mathbf{r}+\mathbf{r}')/2$ and $\mathbf{y} = (\mathbf{r}-\mathbf{r}')/2$. With our "minimally non local" implementation of the Fierz transformations, it is convenient to single-out the local regulator from the potential $V(\mathbf{k},\mathbf{q}) = f_\Lambda(\mathbf{q})V_{\rm c}(\mathbf{k},\mathbf{q})$

$$\begin{aligned}
V(\mathbf{r}',\mathbf{r}) &= \frac{1}{8}\int \frac{d^3k}{(2\pi)^3}\frac{d^3q}{(2\pi)^3} e^{i\mathbf{q}\cdot\mathbf{x}}e^{-i\mathbf{k}\cdot\mathbf{y}}\,f_\Lambda(\mathbf{q})V_{\rm c}(\mathbf{k},\mathbf{q}) \\
&= \frac{1}{8}V_c(i\vec{\nabla}_y, -i\vec{\nabla}_x)\delta^{(3)}(\mathbf{y})I^0_\Lambda(\mathbf{x})\,,
\end{aligned} \tag{20}$$

In the latter equation we have defined

$$I^0_\Lambda(\mathbf{x}) \equiv \int \frac{d^3q}{(2\pi)^3}e^{i\mathbf{q}\cdot\mathbf{x}}f_\Lambda(\mathbf{q}) = \left(\frac{\Lambda}{2\sqrt{\pi}}\right)^3 e^{-\Lambda^2x^2/4} \tag{21}$$

that satisfies $\lim_{\Lambda\to\infty} I^0_\Lambda(\mathbf{x}) = \delta^3(\mathbf{x})$. Beside the spin-orbit term, we consider contributions of the potential that only depend upon the momentum transfer. Thus, their coordinate-space matrix elements are diagonal (the potential is local)

$$V(\mathbf{r}', \mathbf{r}) = \delta^{(3)}(\mathbf{r} - \mathbf{r}') V_c(-i\vec{\nabla}_r) I^0_\Lambda(r) \tag{22}$$

where we have used $\delta^{(3)}(\mathbf{y}) = 8\,\delta^{(3)}(\mathbf{r} - \mathbf{r}')$. We define the coordinate-space version of the potential by applying it to a test function

$$\langle \mathbf{r}'|V|\psi\rangle = \int d^3r V(\mathbf{r}', \mathbf{r})\psi(\mathbf{r})\,. \tag{23}$$

For a local interaction this leads to

$$\langle \mathbf{r}'|V|\psi\rangle = V_c(-i\vec{\nabla}_{r'}) I^0_\Lambda(r') \equiv V_{\mathrm{N^kLO}}(\mathbf{r}') \tag{24}$$

It is then immediate to see that the LO potential reads

$$V_{\mathrm{LO}}(r_{ij}) = \left(C_{20}\mathcal{P}_{(2,0)} + C_{02}\mathcal{P}_{(0,2)}\right) I^0_\Lambda(r_{ij}) \tag{25}$$

As for the NLO, we need to first evaluate

$$-\Delta I^0_\Lambda(r) = \left(\frac{3\Lambda^2}{2} - \frac{r^2\Lambda^4}{4}\right) I^0_\Lambda(r)\,, \tag{26}$$

thus

$$V_{\mathrm{NLO}}(r_{ij}) = \left(E_{02}\mathcal{P}_{(0,2)} + E_{20}\mathcal{P}_{(2,0)}\right)\left(\frac{3\Lambda^2}{2} - \frac{r^2\Lambda^4}{4}\right) I^0_\Lambda(r_{ij})\,. \tag{27}$$

Let us consider the N^{2}LO terms. The tensor requires the computation of

$$-\nabla_\alpha\nabla_\beta I^0_\Lambda(r) = \left[\frac{\Lambda^2}{2}\delta_{\alpha\beta} - \frac{r^2\Lambda^4}{4}\hat{r}_\alpha\hat{r}_\beta\right] I^0_\Lambda(r)\,, \tag{28}$$

The treatment of the spin-orbit term is slightly more involved. From Eq. (20) and Eq. (23), we get

$$\begin{aligned}
\langle \mathbf{r}'|V_{LS}|\psi\rangle &= \frac{1}{8} iE_{LS}/2\epsilon^{\alpha\beta\gamma}(\sigma_1+\sigma_2)^\alpha \int d^3r\,[\nabla^\beta_x I^0_\Lambda(\mathbf{x})][\nabla^\gamma_y \delta^{(3)}(\mathbf{y})]\psi(\mathbf{x}+\mathbf{y}) \\
&= -\frac{1}{8} iE_{LS}/2\epsilon^{\alpha\beta\gamma}(\sigma_1+\sigma_2)^\alpha \int d^3r\,[\nabla^\beta_x I^0_\Lambda(\mathbf{x})]\delta^{(3)}(\mathbf{y})\nabla^\gamma_y\psi(\mathbf{x}+\mathbf{y}) \\
&= -\frac{1}{8} iE_{LS}/2\epsilon^{\alpha\beta\gamma}(\sigma_1+\sigma_2)^\alpha \int d^3r\,[\nabla^\beta_x I^0_\Lambda(\mathbf{x})]\delta^{(3)}(\mathbf{y})\nabla^\gamma_x\psi(\mathbf{x}+\mathbf{y})\,.
\end{aligned} \tag{29}$$

Exploiting the delta function to carry out the integral yields

$$
\begin{aligned}
\langle \mathbf{r}'|V_{LS}|\psi\rangle =& -iE_{LS}/2\epsilon^{\alpha\beta\gamma}(\sigma_1+\sigma_2)^\alpha[\nabla^\beta_{r'}I^0_\Lambda(\mathbf{r}')]\nabla^\gamma_{r'}\psi(\mathbf{r}') \\
& iE_{LS}/2\epsilon^{\alpha\beta\gamma}(\sigma_1+\sigma_2)^\alpha I^0_\Lambda(\mathbf{r}')\frac{\Lambda^2}{2}r'^\beta\nabla^\gamma_{r'}\psi(\mathbf{r}')
\end{aligned}
\tag{30}
$$

Collecting all the above results, the N^{2}LO potential reads

$$
\begin{aligned}
V_{\mathrm{N^2LO}}(r_{12}) =& \left(E_{*0}\mathcal{P}_{(*,0)}+E_{*2}\mathcal{P}_{(*,2)}\right)\left[\left(\frac{\Lambda^2}{2}-\frac{r^2\Lambda^4}{12}\right)\sigma_{12}-\frac{r^2\Lambda^4}{12}S_{12}\right]I^0_\Lambda(r)+ \\
&+\left(E_{00}\mathcal{P}_{(0,0)}+E_{22}\mathcal{P}_{(2,2)}\right)\left(\frac{3\Lambda^2}{2}-\frac{r^2\Lambda^4}{4}\right)I^0_\Lambda(r)-E_{LS}\Lambda^2 I^0_\Lambda(r_{12})\mathbf{L}\cdot\mathbf{S}\,,
\end{aligned}
\tag{31}
$$

where $\mathbf{L}=\frac{1}{2i}(\mathbf{r}\times\vec{\nabla}_r)$ and $\mathbf{S}=\frac{1}{2}(\boldsymbol{\sigma}_1+\boldsymbol{\sigma}_2)$.

In order to implement the potential in QMC, it is convenient to introduce the following set of spin-isospin operator

$$
O^{p=1,7}_{12}=\{1,\sigma_{12},\tau_{12},\sigma_{12}\tau_{12},S_{12},S_{12}\tau_{12},\mathbf{L}_{12}\cdot\mathbf{S}_{12}\}\,,
\tag{32}
$$

By expressing the spin-isospin projection operators in terms of $O^{p=1,4}_{12}$, we can write the two-body potential in a similar fashion as the Argonne potentials

$$
\begin{aligned}
V_{\mathrm{LO}}(r_{12}) &= \sum_{p=1}^{4} v^p_{\mathrm{LO}}(r_{12})O^p_{12} \\
V_{\mathrm{NLO}}(r_{12}) &= \sum_{p=1}^{4} v^p_{\mathrm{NLO}}(r_{12})O^p_{12} \\
V_{\mathrm{N^2LO}}(r_{12}) &= \sum_{p=1}^{7} v^p_{\mathrm{N^2LO}}(r_{12})O^p_{12}\,.
\end{aligned}
\tag{33}
$$

In the above equation we have introduced

$$
\begin{aligned}
v^{p=1-4}_{\mathrm{LO}}(r) &= C^p_S I^0_\Lambda(r) \\
v^{p=1-4}_{\mathrm{NLO}}(r) &= E^p_S\left(\frac{3\Lambda^2}{2}-\frac{r^2\Lambda^4}{4}\right)I^0_\Lambda(r) \\
v^{p=1,3}_{\mathrm{N^2LO}}(r) &= E^p_P\left(\frac{3\Lambda^2}{2}-\frac{r^2\Lambda^4}{4}\right)I^0_\Lambda(r) \\
v^{p=2,4}_{\mathrm{N^2LO}}(r) &= (E^p_P+E^p_*)\left(\frac{3\Lambda^2}{2}-\frac{r^2\Lambda^4}{4}\right)I^0_\Lambda(r) \\
v^{p=5-6}_{\mathrm{N^2LO}}(r) &= E^p_*\frac{r^2\Lambda^4}{4}I^0_\Lambda(r) \\
v^{p=7}_{\mathrm{N^2LO}}(r) &= E^p_*\Lambda^2 I^0_\Lambda(r)\,.
\end{aligned}
\tag{34}
$$

The S, P and mixed (*) parameters are readily expressed in terms of the original low-energy constants. Let us begin from C_S^p, which are fully specified in terms of C_{20} and C_{02}

$$
\begin{aligned}
C_S^1 &= \frac{1}{16}(3C_{20} + 3C_{02}) \\
C_S^2 &= \frac{1}{16}(C_{20} - 3C_{02}) \\
C_S^3 &= \frac{1}{16}(-3C_{20} + C_{02}) \\
C_S^4 &= \frac{1}{16}(-C_{20} - C_{02})\,.
\end{aligned} \tag{35}
$$

The variables E_S^p only depend upon E_{20} and E_{02}

$$
\begin{aligned}
E_S^1 &= \frac{1}{16}(3E_{20} + 3E_{02}) \\
E_S^2 &= \frac{1}{16}(E_{20} - 3E_{02}) \\
E_S^3 &= \frac{1}{16}(-3E_{20} + E_{02}) \\
E_S^4 &= \frac{1}{16}(-E_{20} - E_{02})\,,
\end{aligned} \tag{36}
$$

while E_P^p are functions of E_{00} and E_{22} only

$$
\begin{aligned}
E_P^1 &= \frac{1}{16}(E_{00} + 9E_{22}) \\
E_P^2 &= \frac{1}{16}(-E_{00} + 3E_{22}) \\
E_P^3 &= \frac{1}{16}(-E_{00} + 3E_{22}) \\
E_P^4 &= \frac{1}{16}(E_{00} + E_{22})\,.
\end{aligned} \tag{37}
$$

Finally, for the E_*^p we have

$$
\begin{aligned}
E_*^2 &= \frac{1}{12}(E_{*0} + 3E_{*2}) \\
E_*^4 &= \frac{1}{12}(-E_{*0} + E_{*2}) \\
E_*^5 &= \frac{1}{12}(-E_{*0} - 3E_{*2}) \\
E_*^6 &= \frac{1}{12}(E_{*0} - E_{*2}) \\
E_*^7 &= -E_{LS}\,.
\end{aligned} \tag{38}
$$

Instead of choosing the above spin-isospin structure, in which the spin-isospin projection operators are preserved, an additional Fierz transform can be carried out. This allows to write the potential in the spin-isospin operator basis, although sub-leading regulator artifacts might be introduced. A possible choice, consistent with the one used in Ref. [], follows

$$\begin{aligned}
V_{\mathrm{LO}}(r_{12}) &= (C_0 + C_\sigma \sigma_{12}) I_0^\Lambda(r_{12}) \\
V_{\mathrm{NLO}}(r_{12}) &= (E_0 + E_\sigma \sigma_{12}) \left(\frac{3\Lambda^2}{2} - \frac{r_{12}^2 \Lambda^4}{4} \right) I_0^\Lambda(r_{12}) \\
V_{\mathrm{N^2LO}}(r_{12}) &= (F_0 + F_\tau \tau_{12}) \left[\left(\frac{3\Lambda^2}{2} - \frac{r_{12}^2 \Lambda^4}{4} \right) \sigma_{12} - \frac{r^2 \Lambda^4}{4} S_{12} \right] I_0^\Lambda(r_{12}) \\
&\quad + (E_\tau \tau_{12} + E_{\sigma\tau} \sigma_{12} \tau_{12}) I_0^\Lambda(r_{12}) + F_{\mathrm{LS}} \Lambda^2 I_0^\Lambda(r_{12})\, \mathbf{L} \cdot \mathbf{S}\,.
\end{aligned} \tag{39}$$

Note that the above parametrization can be cast in a form analogous to that of Eq. (33) provided that

$$\begin{aligned}
C_S^1 &= C_0 \\
C_S^2 &= C_\sigma \\
E_S^1 &= E_0 \\
E_S^2 &= E_2 \\
E_P^3 &= E_\tau \\
E_P^4 &= E_{\sigma\tau} \\
E_*^2 &= F_0 \\
E_*^4 &= F_\tau \\
E_*^5 &= -F_0 \\
E_*^6 &= -F_\tau \\
E_*^7 &= F_{LS}\,,
\end{aligned} \tag{40}$$

while all the other parameters vanish.

2.4 Three body force

In Pionless EFT, three-body forces (3NF) are promoted to LO to avoid the Thomas collapse of $A \geq 3$ systems. The collapse only appears for ($S = 1/2$, $I = 1/2$), as this is the only channel that allows a antisymmetric wave function for three-particles on the same position in space. Therefore, a 3NF capable of preventing the collapse reads

$$V_{ijk}^{LO} = D_1 \sum_{i<j<k} \sum_{cyc} \mathcal{P}_{(1,1)}^{(ijk)} \tag{41}$$

When applied to an antisymmetric wave function, the effect of the 3NF above is identical to those in which $\mathcal{P}^{(ijk)}_{(1,1)}$ is replaced by one of the following six operators

$$1, \sigma_{ij}, \tau_{ij}, \sigma_{ij}\tau_{ij}, (\sigma_i \times \sigma_j)\cdot\sigma_k, (\tau_i \times \tau_j)\cdot\tau_k \tag{42}$$

When the local regulator $f_\Lambda(q)$ is introduced, 3NF are affected by regulator artifacts analogous to those of the two-body case. In our QMC calculations we adopted the simple form

$$V^{LO}_{ijk} = D_1 \sum_{i<j<k} \sum_{cyc} \left[f_\Lambda(q_{ik}) f_\Lambda(q_{jk})\right] \tag{43}$$

as in the limit $\Lambda \to \infty$ is equivalent to

$$V^{LO}_{ijk} = D_1 \sum_{i<j<k} \sum_{cyc} \left[f_\Lambda(q_{ik}) f_\Lambda(q_{jk}) \mathcal{P}^{(ijk)}_{(1,1)}\right] \tag{44}$$

up to sub-leading contributions.

2.5 Fitting observables

Since EFT($\not\pi$) is equivalent to an effective range expansion (ERE) in partial waves, the ERE parameters are a natural choice to fit the LECs of the theory. However, any combination of observables or phase shifts can be used as long they depend on the correct low energy quantities. The identification of each parameter is listed in tab. The correspondence between each LECs and the observable according to the angular momentum and the momentum power in which they appear is displayed in Table 1. Note that the demotion of part of the NLO potential leaves the first two order of the theory acting only in S-wave as described in [Effective Field Theory of Nuclear Forces by B. Van Kolck]

$$\begin{array}{l} <\psi|V(p^2,p'^2)|\psi'>= \\ \int d|p|d|p'|V(p^2,p'^2)\int d\hat{p}Y_{lm}1\int d\hat{p'}Y_{l'm'}1 = \\ \frac{1}{(4\pi)^2}\int d|p|d|p'|V(p^2,p'^2)\delta_{(l=0)}\delta_{(l'=0)}\,. \end{array} \tag{45}$$

On the other hands the terms $p\cdot p'$, $\mathbf{L}\cdot\mathbf{S}$ and S_{12}, which act on partial waves with angular momentum l> 1, appear to be at N^{2}LO along with the shape parameter of the S-wave.

It has to be stressed that each sub-leading operator has to be treated in the corresponding order of perturbation theory. The LECs of any specific order have to be simultaneosly fitted, without changing the parameters of the lower orders that were previously determined. To this aim, a counter-term for each leading order operator has to be introduced to ensure that the observable is not changed when perturbative corrections are included.

LEC	Fit observable	Source
$C_{(2,0)}$	$a^{(l=0)}$(d)	5.41 fm
$C_{(0,2)}$	$a^{(l=0)}$(nn)	-18.63 fm
$E_{(2,0)}$	$r_0^{(l=0)}$(d)	2.75 fm
$E_{(0,2)}$	$r_0^{(l=0)}$(nn)	2.77 fm
$E_{(0,0)}$	$a^{(l=1)}$(d)	phaseshifts
$E_{(2,2)}$	$a^{(l=1)}$(nn)	phaseshifts
$D_{(1,1)}$	B(^{3}H)	-8.482 MeV

Table 1: Table of the observable used to fit EFT($\not\pi$) LECs.

2.6 Work steps

In this subsection we describe the steps we intend to take for the realization of this work, which is mainly aimed at i) Providing a more accurate comparison against Lattice-QCD data, ii) Assessing the issue of unbound ^{16}O.

A) ^{4}He with perturbative NLO. This is the rigorous way of including NLO in EFT($\not\pi$) . Betzalel and Konig have preliminary (unpublished yet) results for the NLO potential in the form of Eq. (14). They showed that EFT($\not\pi$) NLO is not renormalizable and the observables have an unexpected residual cut-off dependence for nuclei larger than ^{4}He. We want to check whether this issue also appears when using local regulators. We plan to proceed as follows:

- Fit LO parameters non perturbatively.
- Fit NLO parameters perturbatively using scattering lengths and effective ranges of S-waves adding counterterms to preserve the observables fitted at LO.
- Calculate the perturbative NLO correction to the ^{4}He binding energy and, eventually ^{16}O, although we are well aware that perturbative correction are unlikely to bind ^{16}O.
- Benchmark MC and SVM results with the same interaction.

A.1) ^{4}He with non-perturbative NLO. We want to understand whether we can reproduce the results of [ArXiv:1712.10246], despite this approach potentially leads to essential regulator dependencies. Indeed, in order to avoid the Wigner bound in the effective rage, the cut-off should be kept finite (λ <4 fm^{-1}). We intend to proceed as follows:

- Simultaneous non-perturbative fit of LO and NLO parameters.
- Calculate the energy of ^{4}He and ^{16}O and compare it with the one of [ArXiv:1712.10246].

B) LO in p-wave systems We would like to study P-wave nuclear systems using S-wave potentials. It might be not possible to describe fermionic systems of in relative P-wave using only S-wave contact interactions. However, to the best of our knowledge there is no decisive proof of this statement. We plan to calculate the binding energy of three neutrons (nnn) and ^{5}He and determine whether for certain values of the LECs these systems are bound. If we find that these P-wave systems are always unbound when using S-waves interactions alone, this could indicate that P-waves should be promoted to correctly describe A> 4 nuclei. More specifically, we plan to

- Study the correlation between the binding energy of the three-neutron system with $a_0^{(nn)}$.
- Calculate the ^{5}He energy with using different combinations two-body and three-body LECSs.

C) Promote P-waves at leading order. If we show that we can not bind P-wave particles with S-wave interaction, it might be not possible to bind ^{16}O with LO and perturbative NLO. Therefore, we are justified to promote the N^{2}LO P-wave contributions to LO to correctly describe the binding energies of $A > 4$ nuclei. The operators to be promoted are all the P-wave N^{2}LO contributions of Eq. (39) with the exception of $E_{*0}(\sigma_1 \cdot q)(\sigma_2 \cdot q)\mathcal{P}_{(*,0)}$, which connects S- and D-wave channels.

- Simultaneous non-perturbative fit of LO and N^{2}LO potential (the full one or only those proportional to $P_{(0,0)}$ and $P_{(2,2)}$).
- Calculate the energies of ^{4}He, nnn, and ^{5}He and compare them to those computed in the previous steps.

D) Perturbative correction of excited states of ^{16}O. We want to understand if it possible to switch the ground-state of a many-body system with an excited state using a perturbative interaction. It is clear that the perturbative expectation value of the energy computed on a four-^{4}He state cannot be lower than four times the binding energy of a single ^{4}He. However, it might be possible that the energy of an excited non-clustered state (a resonance) turns out to be more bound than four-^{4}He when perturbative NLO is introduced. We should devise a QMC strategy to tackle this problem.